

## 7.5 Visualising time series data ↕

Before discussing how to visualise time series data, it's worth remembering some key definitions.

### Time series data

A collection of observations gathered through repeated measurements, collected at different points in time. They are normally ordered chronologically.

### Cross-sectional data

This is a collection of observations taken at a single point in time. In general, time series data should be visualised using line charts. Line charts emphasise sequential flow, with the line connecting the consecutive data points visualising any trend, variability or seasonality in the data.

### Trend

A trend is a pattern in the values that is observable in the data over several periods of time.

### Trendlines

A trendline assists in visualising a trend over time. They are useful for visualising a long-term trend in the data. However, it is important to be aware that the whole range of data is used to fit the trendline. For time series data, it is often advisable to smooth the data in a way that:

- only uses observations known up to the time at which the smoothing estimate is calculated
- does not use observations from the distant past.

### Variability

In the case of time series data, variability is the difference in values from one period to another.

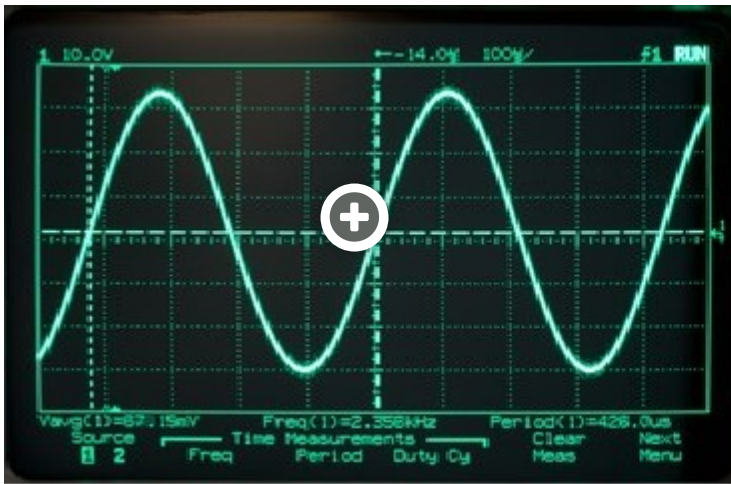
### Seasonality

Seasonality is a pattern that recurs periodically in the data over a regular time period, and so the pattern becomes predictable. Seasonality can refer to any regular time interval, including hourly, daily, weekly, etc. It can be difficult to clearly identify in charts that provide a chronological display of all of the data (e.g. data extending over several years). Therefore, it is often better to use multiple lines in a chart, with each line representing one specific time interval, such as one year.

When comparing individual values of one or more variables over a time period, it may be useful to use a column chart.

## Viewing data at different temporal frequencies

Click on the images below to learn more about temporal frequency and aspect ratio.



## Highlighting patterns in time series data

There are different types of patterns that can be highlighted in time series data.

### Moving average

This can be calculated by averaging the last “x” values observed over an x-period. This means that observations prior to the x-period and future observations are not included in the moving average calculation.

### Trellis display

This can be used to vertically or horizontally display a series of charts showing the effects of different times series charts (e.g. different moving averages), developed from the same time series data. The charts are the same size and have the same scales and formatting. The only difference is the data that is displayed in each chart.

## Rearranging data for visualisation

A time series dataset comprises rows of data, each of which represents an observation. Each observation is taken at a specific point in time and provides the values of the set of variables. This data can be organised in many ways, particularly if at least one of the variables is categorical.

Data may be provided/arranged in either stacked or unstacked forms:

- *Stacked data*

The data is organised so that the values for all groups are in a single column. The column labels identify the group.

- *Unstacked data*

The data is organised so that each column contains observations from one group/category. Therefore, the values for a categorical variable are stored in separate labelled columns. The columns contain the observed variable values which correspond to the respective group/category.

Differing data arrangements facilitate different types of visualisations. Therefore, it may be necessary to transform data from stacked form to unstacked form and vice versa.

A *sparkline* is a particular type of line chart that provides a visual representation of data. It shows a trend but does not show magnitude. Sparklines are very small (therefore take up little space); they are placed in a cell in a spreadsheet. For best effect/greatest impact, they should be placed near to the relevant data. They do not have axes or labels. They can be used to explore and compare interesting patterns across a categorical variable in time series data. They are easy to produce and provide an efficient way of displaying basic time series information.

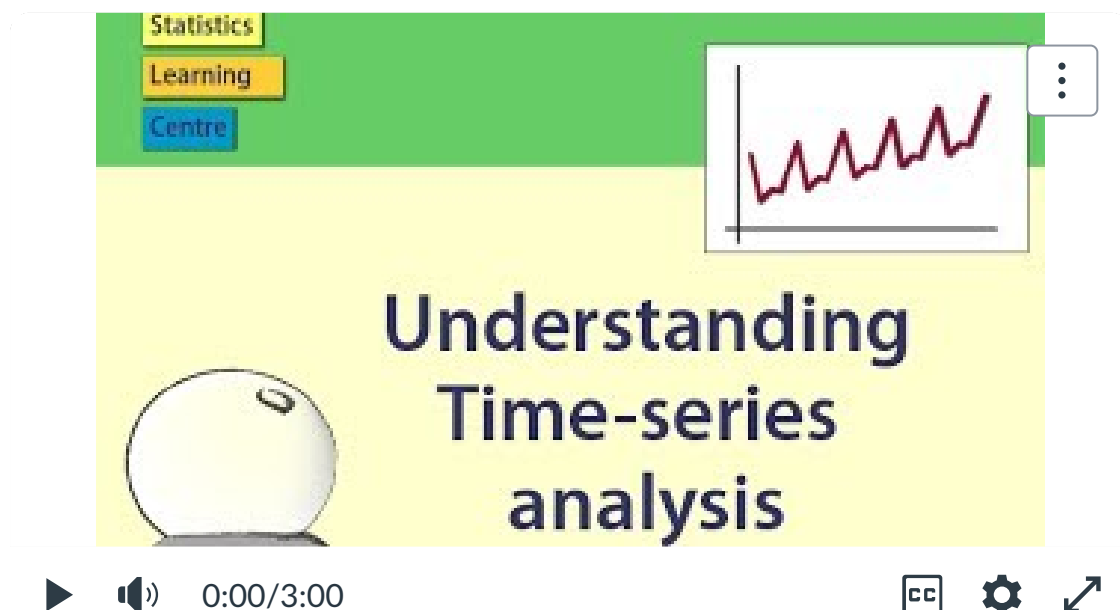
Box and whisker charts (see Module 5) may also be useful for visualising time series data and can be constructed from unstacked data.

## □ Watch: Introducing time series analysis and forecasting [2.59 mins]

This video is useful as it explains what a time series is and introduces concepts such as trends, seasonality and cycles.

As you watch, consider:

- How can time series analysis be used to explore and reveal data insights?
- At what stage of data analysis time series is analysis important?



Source: © Dr Nic's maths and stats (2014).

## □ Activity: Comparing the speed of interest rate hikes (1988–2023)

Reflect on this webpage about comparing interest rate increases over time.

As you look at the visualisations, consider and make notes on:

- how you could improve the first visualisation
- whether you would have used another approach to represent the data
- what techniques you would use, and why.

Explain your reasoning and bring your notes to the workshop.

Ross, J., & Lam, S. (2023, May 7). [Comparing the speed of interest rate hikes \(1988-2023\)](https://www.visualcapitalist.com/interest-rate-hikes-1988-2023/). [Visual Capitalist](https://www.visualcapitalist.com/interest-rate-hikes-1988-2023/).